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Stage-dependent abnormalities induced by the fungicide triadimefon in the mouse☆

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ABSTRACT

Aim of this work is the study of abnormalities induced by the triazole triadimefon (FON) administered to pregnant mice at E8, E9, E10, E11 or E12. Pregnant CD-1 mouse were gavaged with FON 500 mg/kg at the selected stages and sacrificed at term and fetuses morphologically examined and processed for visceral and skeletal analysis. Administration of FON on E8, E10–E12 resulted in fetuses with cleft palate (E8 39% and E12 24% representing the peak of sensitivity, in E8 fetuses associated to severe skull basis abnormalities). Other cranial malformations (fusions abnormalities or agenesis of bones) were observed in E8–E10 groups (E8 the most sensitive with 96% of malformed fetuses). Cardiovascular abnormalities were observed in a stage dependent manner at E8–E10 (22.2, 3.8, 7.8%). As far as craniofacial malformation is concerned, we propose that FON acts on two different stages, involved in early and late craniofacial formation.

1. Introduction

Azoles are antifungals widely used as agrochemicals against mildews and rusts of cereal grains, fruits, vegetables, or clinically against topical or disseminated candidiasis and coccidioidal meningitis (the clinical dosage usually suggested is 100–400 mg/day, but larger doses are recommended for candidaemia, cryptococcal or coccidioidal meningitis; the route of administration is dermal, oral or intravenous injection). Azole derivatives act via the inhibition of CYP enzymes involved during the fungal wall biosynthesis (CYP 51) [1,2]. Depending on their chemical properties, they are used both in agriculture and in clinical therapy. Side effects of azoles (mainly hepatotoxic effects) are mostly due to the inhibition of mammalian CYPs [3] and have been related to affected steroidogenesis [4]. All the tested azoles are teratogenic in experimental animals: dose-related effects have been reported after *in vitro* exposure of postimplantation rodent embryos [5–14] or after exposure of pregnant rats and mice [15–19]. All the studied azoles induce the same malformation pattern: the reported embryonic abnormalities were related to the branchial apparatus (abnormal or fused branchial arches) [5–14]. After external and skeletal analysis of CD-1 mouse fetuses exposed *in utero* on day 8 (E8) to 300 mg/kg *per os* of the agrochemical triadimefon (FON) and examined at term of gestation, cleft palate has been observed in 23% of samples, 87% showed cranio-facial defects reconducible to alteration of skeletal derivatives originating from the embryonic branchial apparatus (middle ear, squamosal, zygomatic, alisphenoid, styloid, pterygoid, mandible abnormalities and fusion of skeletal elements) and axial defects (fusion, homeotic respecification and duplication of segments). Interestingly, 77% of examined treated fetuses showed ectopic cartilage at the level of upper jaw [18]. Similar results were obtained by treating *per os* pregnant rats on day 9.5 *post coitum* with 500 mg/kg FON [19].

Tiboni and Giampietro [17] determined phase specific effects of the clinically used Fluconazole by treating CD-1 mice with single oral dose of 700 mg/kg on gestational day 8, 9, 10, 11, or 12. Gestational day 10 was identified as the phase of maximal sensitivity as indicated by the postimplantation loss and for induction of cleft palate. Middle ear abnormalities were detected after exposure on gestational day 8, while limb defects were characteristic of the group exposed on day 10 of gestation. In addition, the authors identified a dose–response relationship, investigated on gestational day 10, showing a clear dose–response in the induction of postimplantation loss and cleft palate.

The aim of this study is to verify if a stage-specificity relationship is detectable also after *in utero* FON exposure, and to expand knowledge on FON-related effects. For this purpose, pregnant CD-1 mice at different gestational stages (E8, E9, E10, E11 or E12) have been treated with a single oral administration of 500 mg/kg FON, and fetuses examined at term of gestation for external, visceral and skeletal morphology.

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Table 1
Maternal and fetal parameters.

	CON	FON 8	FON 9	FON 10	FON 11	FON 12
Pregnant dams	13	9	13	9	8	12
mbw change (0–18)	34.69 ± 4.87	27.78 ± 4.74	27.78 ± 9.65	31.0 ± 04.72	31.75 ± 8.03	28.33 ± 5.72
Litter weight	24.08 ± 2.60	19.56 ± 7.13	18.67 ± 7.00	23.00 ± 2.87	23.50 ± 5.18	21.00 ± 5.73
mbw change without litter	10.62 ± 4.35	8.22 ± 5.19	9.11 ± 3.14	8.00 ± 2.60	8.25 ± 3.11	7.33 ± 2.66
Implantations	13.62 ± 1.56	11.56 ± 4.03	11.56 ± 3.91	14.33 ± 1.50	14.13 ± 3.14	12.83 ± 3.43
Dead fetuses	0.08 ± 0.28	0.11 ± 0.33	0.22 ± 0.67	0.22 ± 0.44	0.13 ± 0.35	0.00 ± 0.00
Resorptions	0.08 ± 0.28	1.33 ± 0.58	0.89 ± 1.17	0.75 ± 0.89	0.75 ± 0.71	1.50 ± 1.38
Live fetuses	13.38 ± 1.80	11.00 ± 4.12	10.44 ± 3.84	13.56 ± 1.42	13.25 ± 3.15	11.33 ± 3.93
Postimplantation loss (%)	1.19 ± 2.92	4.95 ± 6.91	10.14 ± 10.88	5.89 ± 7.03	6.54 ± 4.80	12.81 ± 11.73[a]
Fetal weight	1.44 ± 0.11	1.43 ± 0.14	1.41 ± 0.09	1.33 ± 0.09	1.39 ± 0.07	1.47 ± 0.14

Data expressed as means and standard deviations of the litter means. mbw, maternal body weight.
[a] $p < 0.05$ vs. control.

2. Materials and methods

Virgin CD-1 mice (Charles-River, Calco, Italy), housed in a thermostatically maintained room ($T = 22 \pm 2\,^{\circ}C$; relative humidity $55 \pm 5\%$; light cycle from 7:00 AM to 7:00 PM) with food (4RF21; Charles River Laboratories) and tap water *ad libitum*, were caged overnight with males of proven fertility. The day of the vaginal plug was considered as E0. 10 pregnant mice/group were orally treated by gavage with 500 mg/kg FON (Fluka, Milan, Italy; dissolved in a solution containing 10% acetone in corn oil) on E8, E9, E10, E11 or E12 at 10:00 AM. The dosage was selected on the basis of previous range-finding tests in order to obtain almost 100% of *fetuses* with head abnormalities after dam treatment on E8. Controls were dosed with vehicle alone on E8–E12. Females were checked for signs of toxicity after the treatment until the sacrifice and weighted on E0, at the time of treatment and at term of gestation.

2.1. Analysis at term

After dam sacrifice by CO_2 asphyxia, the entire uterine content was weighed and total implantation, live and dead fetuses, early and late resorptions were recorded. The postimplantation loss index [((total implantation − live fetuses)/total implantation) × 100] was calculated. Live fetuses were then explanted from the uterus, morphologically examined, and weighed. The examined fetuses were processed for visceral examination according to the free-hand razor blade sectioning method proposed by Wilson [20] or for the double skeletal staining of bone and cartilage, carried out according to the method described by Kimmel and Trammell [21], partially modified as previously described [22]. The osseous and cartilaginous tissues were stained using respectively alizarin red and alcian blue reagents. After staining, fetuses were analyzed under a dissecting microscope and the abnormalities were recorded.

2.2. Statistical analysis

Statistical analysis (Student's *t*-test) was performed on mean and standard deviation of maternal body weight (mbw) change (E0–18), litter weight, mbw change without litter (mbw change E0–18 – litter weight) and on mean and standard deviation of postimplantation loss index and fetal weight (using litter as experimental unit). The incidence of fetuses with defects was analyzed both by using fetus and litter as experimental unit. The percentage of fetuses and litters affected was analyzed using the Chi-square test. The level of significance was set at $p < 0.05$.

3. Results

No significant signs of maternal toxicity were detected in the treated groups, as shown by clinical signs and by the weight gain after treatment, even if a not significant trend of weight reduction (without litter) was observable (Table 1). FON exposure induced a general increase of postimplantation loss index, statistically significant on E12 (Table 1).

As far as the external examination is concerned, except for E9 group, FON exposure induced cleft palate in fetuses (Table 2). The timing of exposure was critical to determine this malformation: the peak sensitivity for this malformation was after treatment on E8. Interestingly only 1.1% of fetuses exposed on E9 showed this malformation, while the percentage progressively increased from E10 to E12. At the visceral examination, a stage-related significant increase of cardiovascular abnormalities was observed at E8–E10 (respectively 2.2 and 3.8% of great vessels abnormalities at E8 and E9, 3.1 dextrocardia and 4.7 intraventricular septal defects at E10) treatment dosage groups (Table 2).

After skeletal double staining, the morphological analysis of cranial basis bones revealed a complete disruption of bone architecture only in the group treated with FON on E8: fusion of elements

Table 2
External and visceral examination of fetuses: % of affected fetuses/litters. Statistics was done only on bold parameters.

	CON	FON 8	FON 9	FON 10	FON 11	FON 12
Pregnant dams	13	9	13	9	8	12
Total externally examined fetuses	174	99	94	122	106	126
Cleft plate	**0/0**	**39.4**[aa]/**88.9**[aa]	**1.1**[bb]/**0.08**[bb]	**16.4**[aa,bb,cc]/**33.3**[aa,bb,cc]	**15.1**[aa,bb,cc]/**75.0**[bb,cc,dd]	**23.8**[aa,b,cc]/**33.3**[aa,bb,cc,ee]
Total viscerally examined fetuses	85	45	53	64	52	63
Total malformed at visceral examination (cardiac defects)	**0/0**	**2.2/11.1**[a]	**3.8/15.4**[aa]	**7.8**[a]/**44.4**[aa,bb,cc]	**0/0**[b,cc,dd]	**0/0**[b,cc,dd]
Great vessels abnormalities	0/0	2.2/11.1	3.8/15.4	0/0	0/0	0/0
Interventricular septal defects	0/0	0/0	0/0	4.7/11.1	0/0	0/0
Dextrocardia	0/0	0/0	0/0	3.1/22.2	0/0	0/0

[a] $p < 0.05$ vs. control.
[aa] $p < 0.01$ vs. control.
[b] $p < 0.05$ vs. FON 8.
[bb] $p<0.01$ vs. FON 8.
[c] $p < 0.05$ vs. FON 9.
[cc] $p < 0.01$ vs. FON 9.
[d] $p < 0.05$ vs. FON 10.
[dd] $p < 0.01$ vs. FON 10.
[e] $p < 0.05$ vs. FON 11.
[ee] $p < 0.01$ vs. FON 11.

Table 3
Examination of fetal head after double staining: % of affected fetuses/litters. Statistics was done only for bold parameters.

	CON	FON 8	FON 9	FON 10	FON 11	FON 12
Pregnant dams	13	9	13	9	8	12
Total double staining examined fetuses	89	54	41	58	54	63
Malformed fetuses (total)	**0/0**	**96.3**^{aa}/**100**^{aa}	**41.5**^{aa,bb}/**87.5**^{aa,b}	**10.3**^{aa,bb}/**11.1**^{aa,bb,cc}	**0**^{bb,c,d}/**0**^{bb,cc,d}	**0**^{bb,c,d}/**0**^{bb,cc,d}
Fusions	**0/0**	**74.1**^{aa}/**87.5**^{aa}	**36.6**^{aa,bb}/**62.5**^{aa,bb}	**0**^{bb,cc}/**0**^{bb,cc}	**0**^{bb,cc}/**0**^{bb,cc}	**0**^{bb,cc}/**0**^{bb,cc}
Squamosus/tympanic ring		20.4/62.5	7.3/12.5			
Squamosus/zygomatic		53.7/62.5	7.3/25.0			
Mandible/zygomatic		5.6/25.0				
Squamosus/pterygoid		40.7/75.0	2.4/12.5			
Squamosus/alysphenoid		20.4/37.5				
Alysphenoid/tympanic ring		5.6/37.5				
Basioccipital/basisphenoid		9.3/12.5				
Pterygoid/tympanic ring		25.9/75.0				
Squamosus/parietal		14.8/62.5	4.9/12.5			
Mandible/tympanic ring		7.4/25.0	9.8/25.0			
Squamosus/mandible		14.8/50.0	14.6/25.0			
Abnormalities	**0/0**	**66.7**^{aa}/**100**^{aa}	**43.9**^{aa,b}/**75.0**^{aa,bb}	**10.3**^{aa,bb,cc}/**11.1**^{aa,bb,cc}	**0**^{bb,cc,d}/**0**^{bb,cc,d}	**0**^{bb,cc,d}/**0**^{bb,cc,d}
Tympanic ring		22.2/75.0				
Squamosus		27.8/75.0	24.4/37.5	10.3/11.1		
Alysphenoids		22.2/50.0				
Pterygoids		11.1/37.5				
Mandible		22.2/62.5	2.4/12.5			
Extra ossificated element	**0/0**	**0/0**	**34.1**^{aa,bb}/**50.0**^{aa,bb}	**0**^{cc}/**0**	**0**^{cc}/**0**	**0**^{cc}/**0**
Ectopic cartilage	**0/0**	**50.0**^{aa}/**62.5**^{aa}	**0**^{bb}/**0**	**0**^{bb}/**0**	**0**^{bb}/**0**	**0**^{bb}/**0**
Agenesis	**0/0**	**5.6**^{a}/**25.0**^{aa}	**0**^{b}/**0**^{bb}	**10.3**^{aa,cc}/**11.1**^{aa,cc}	**0**^{b,dd}/**0**^{bb,dd}	**0**^{b,dd}/**0**^{bb,dd}
Squamosal zygomatic process				10.3/11.1		
Tympanic ring		5.6/25.0				

^{a} $p < 0.05$ vs. control.
^{aa} $p < 0.01$ vs. control.
^{b} $p < 0.05$ vs. FON 8.
^{bb} $p < 0.01$ vs. FON 8.
^{c} $p < 0.05$ vs. FON 9.
^{cc} $p < 0.01$ vs. FON 9.
^{d} $p < 0.05$ vs. FON 10.
^{dd} $p < 0.01$ vs. FON 10.
^{e} $p < 0.05$ vs. FON 11.
^{ee} $p < 0.01$ vs. FON 11.

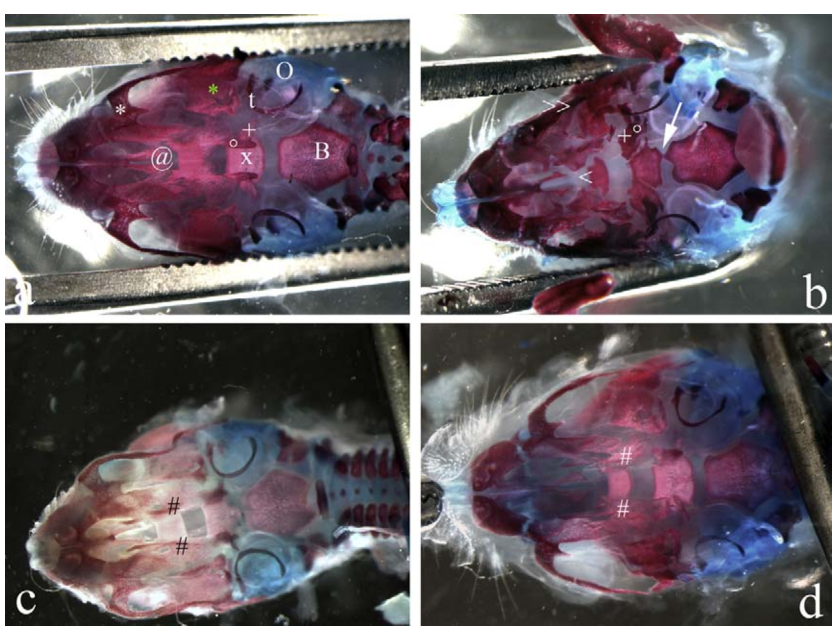

Fig. 1. View of cranial basis skeletal elements in a control fetus (a) or in fetuses exposed to FON at different gestational ages (E8, b; E10, c; E12, d). In the control sample (a) note: B, basioccipital bone; x and +, basisphenoid and alisphenoid bones;°, pterygoid process; @, palatine; black * and white *, basal portion of squamosal and maxillary bones; O, otic capsule; t, tympanic ring. In E8 treated fetus (b) note the large cleft palate (>), the abnormal position and shape of alisphenoid bones (+), in reference to the basisphenoid, the abnormal pterygoid process (°). Note also the upper jaw ectopic cartilage (») joining the basal cartilages and the abnormal Meckel's cartilage reaching the middle ear fused to the ear ossicles and the fusion between basishenoid and basioccipital bones (→). In later treated fetuses (c and d) note the unfused palatal shelves (#). Magnifications: 8×.

and morphological abnormalities were associated with palatal shelve malformations and cleft palate (Table 3 and Fig. 1b). On the contrary, the totality of cleft palate observed after treatment on E10, E11 or E12 was related to normally shaped but not fused palatal shelves (Fig. 1c and d).

Other craniofacial specific and severe abnormalities were observed after skeleton staining in 96.3% of the fetuses explanted from mothers treated with FON on E8 (Table 3 and Fig. 2c and d). The abnormalities were abnormal shape, reduction, agenesis, and fusion of skeletal elements. Ectopic cartilage at the level of upper jaw was observed, visible as ectopic cartilaginous fragments or as a large element sometimes joining the upper and lower jaw together (Fig. 2c and d). Jaw and temporal abnormalities were also observed after treatment with FON on E9 or E10. A stage-dependent decrease of effects was noted both as percentage of affected fetuses and for the severity of malformations (fusion of elements coupled with squamosal abnormalities in 41.5% of fetuses exposed on E9; abnormal shape of squamosus in 10.3% of fetuses exposed on E10) (Table 3 and Fig. 2e–h).

4. Discussion

In the present work, we demonstrate a stage-specific trend in the occurrence of craniofacial malformations after single

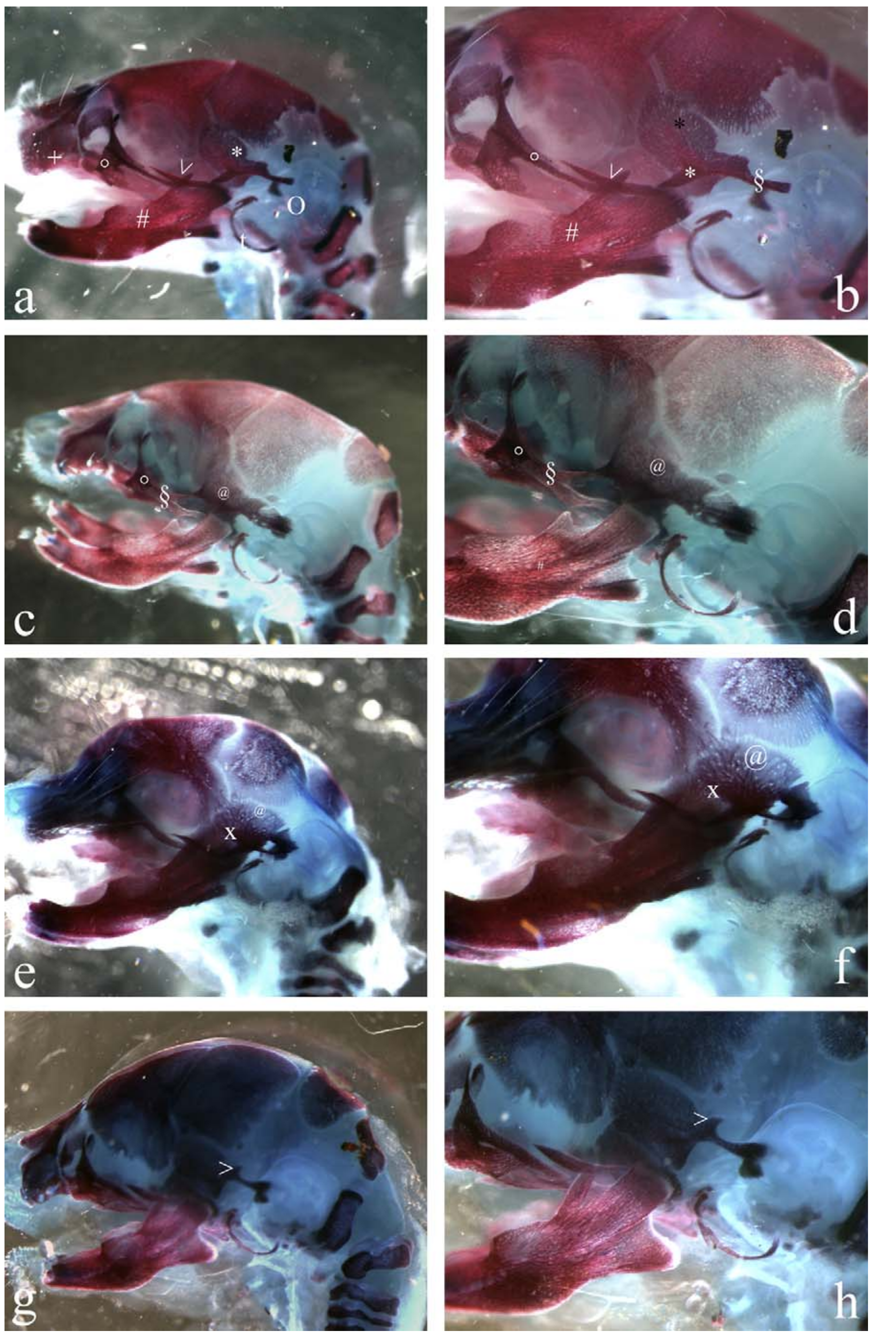

Fig. 2. Lateral appearance of the normal skull at term of gestation (a and b). (a) Maxillary structures: +, premaxillary; °, maxillary; >, zygomatic; black *, squamosal bones. No cartilaginous elements are visible in this region. At the mandible level, the dental bone (#) is recognizable. The tympanic ring (t) is clearly visible near the otic capsule (O). (b) Magnification of (a). Note the normal structure of the squamosal bone (black *) with the retro tympanic (§) and zygomatic (white *) processes. °, zygomatic process of the maxillary bone; #, dental bone; >, zygomatic bone. (c and d) Lateral view of a E8 FON fetus. Note the fusion of the upper jaw bones (§), the ectopic cartilage (°), the abnormal mandible (#) and abnormal squamosus (@). (e and f) Lateral view of a E9 FON fetus. Note the squamosal abnormalities (@) and the fusion between squamosus and mandible (x). (g and h) Lateral view of a E10 FON fetus. Note the squamosal abnormalities (>). Magnifications: a, c, e and g: 8×; b, d, f and h: 20×.

dosage of FON on E8, E9, E10, E11 or E12. The reported craniofacial abnormalities were cleft palate and disruption of skull elements.

The complex malformative pattern (including cleft palate) described in the present work after exposure to FON early in pregnancy (E8) could be explained as the result of the perturbation of migrating neural crest cells.

As demonstrated by Jiang et al. [23], neural crest contribution is fundamental for the mammalian skull morphogenesis. In particular, in mammals, the nasal, frontal, inter-parietal, alysphenoid bones and ear ossicles are totally originated from neural crest contribution, while hyoid and temporal bones have a double origin from mesoderm and neural crest population. As far as the temporal bone is concerned, the squamosal region is a neural crest derivative. The observation of the skull embryogenesis allowed to show a spatio- and temporal-specification of elements involved in the morphogenesis of the different skull elements: the first neural crest cells colonizing the first branchial arch are instructed by the branchial mesoderm to differentiate into the mandibular process, the later migrating elements, by contrast, are progressively induced to differentiate the maxillary process and the squamosal region [24]. In mammals the neural crest derivatives of the mandibular and maxillary processes originate the jaw elements, the malleus, the Meckel's cartilage and the incus [25].

In the present work, a good correlation between stage of migrating neural crest cells at the time of treatment and abnormal skeletal elements at term is possible: the most serious alterations can be identified in the groups where no crest cells already have been instructed to have a determined fate (E8), while no effects are visible after migration and specification of crest cells occurred (E11–E12).

As far as the secondary palate is concerned, a double peak was observed at E8 and E12. The skeletal analysis, however, revealed that this external malformation was associated to a complete disruption of bone architecture only in the group treated with FON on E8, while it resulted as the consequence of unfused normally shaped palatal shelves in E10–E12 group.

As described by McBratney-Owen et al. [26] the cranial base of the mouse develops in a stage-dependent manner from E8 till E16, when the formation of the condrocranium is complete. Neural crest cells destined for the anterior cranial base migrate at E8, while mesoderm derived tissues originate the posterior part of the cranial base, chondrifying at later stages [23].

Around at E11.5, in mice, secondary palate primordia appear organized on the medial wall of the maxillary processes, then secondary palate development involve the growth, elevation, medial elongation and midline fusion of already formed palatal shelves [27].

The cleft palate described in the present work after treatment with FON of pregnant mice during the early developmental stage (E8) could be due to a perturbation of the palate primordia morphopgenesis, while the same external malformation reported after dosage at later developmental stages (E10–E12) could be solely the result of the failure of closure of bilateral palatal shelves in the midline.

Cleft defects as well as craniofacial and limb defects have been described in experimental animals after the *in utero* or *in vitro* exposure to a number of azole fungicides, and for some of them dose-dependency and stage-specificity have been described, even if the specific mechanism of action has never been demonstrated [28,29]. Previously reported experimental data [8–14] strongly suggest that the inhibition of mammalian CYP enzymes could be the mechanism accounting for the teratogenic effects of these molecules. In particular, the inhibition of CYP26 (the enzyme involved in the catabolism of retinoic acid) has been suggested as the mechanism related to neural crest cell migration alterations [8–14]. To explain the multiple effects on FON during early and later stages of palatogenesis, however, the inhibition of several CYP enzymes involved in palatogenesis (including those involved in cholesterol synthesis) has to be taken in account.

Finally, cardiovascular defects have been reported after the visceral examination in at stage-related manner at E8–E10. Such effects are a novelty for the azole literature on experimental models and were not reported by Tiboni and Giampietro [17] in their work on stage-specific effects of a single exposure to Fluconazole. The cardiovascular effects obtained in the present work could be related to a disruption of cardiac neural crest cell migration involved in the outflow septation. To confirm these data and this hypothesis, further investigations are necessary.

To explain the different peaks of sensitivity and effects observed in our present work if compared to the previously published data on Fluconazole [17], considering the fact that different azole-derivatives show different inhibitory properties on a variety of CYP isoforms, the possibility that different azoles can selectively inhibit specific members of CYP family has to be considered. A different pharmacokinetic pathway of FON and Fluconazole could also alternatively be suggested to explain the not completely concordant data between FON and Fluconazole works. These speculative hypotheses, however, need further *ad hoc* experiments.

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